

3 **(a)** resonance B1 [1]

(b) $Pt = mc \Delta\theta$ C1

$750 \times 2 \times 60 = 0.28 \times c \times (98 - 25)$ C1

$c = 4400 \text{ J kg}^{-1} \text{ K}^{-1}$ A1 [3]

(use of $\Delta\theta = 73 + 273$ max. 1/3)

(use of $t = 2 \text{ s}$ not 120 s max. 2/3)

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- (c) e.g. some microwave leakage from the cooker
 e.g. container for the water is also heated
(any sensible suggestion)

B1 [1]

4 (a) (i) $F_E = Q_1 Q_2 / 4\pi\epsilon_0 r^2$
 $= 8.99 \times 10^9 \times (1.6 \times 10^{-19})^2 / (2.0 \times 10^{-15})^2$
 $= 58 \text{ N}$

C1

A1 [2]

(ii) $F_G = Gm_1 m_2 / r^2$
 $= 6.67 \times 10^{-11} \times (1.67 \times 10^{-27})^2 / (2.0 \times 10^{-15})^2$
 $= 4.7 \times 10^{-35} \text{ N}$

C1

A1 [2]

- (b) (i) force of repulsion (much) greater than force of attraction
 must be some other force of attraction
 to hold nucleus together

B1

M1

A1 [3]

(Do not allow if $F_G > F_E$ in (a) or one of the forces not calculated in (a))

- (ii) outside nucleus there is repulsion between protons
either attractive force must act only in nucleus
or if not short range, all nuclei would stick together

B1

B1 [2]